Program : **Diploma in Instrumentation Engineering**	
Course Code : **4083**	Course Title: **Process Control Instrumentation**
Semester : **4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L: 3 T: 1 P: 0)**	Periods per semester: **60**

Course Objectives:

- To introduce the concepts of Process and Process control.
- To illustrates various elements in a process control loop and controller tuning.
- To design simple alarm circuits
- To explain functional elements of process control loop and various communication methods in process control applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Classification and types of sensors		Sensors and Transducers	3
Characteristics of instruments		Electronic Instrumentation	2
Control system concepts		Control Engineering	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concepts of automatic Control with examples.	15	Understanding
CO 2	Illustrate controller tuning methods.	14	Applying
CO 3	Explain the concepts of final control operation.	14	Understanding
CO 4	Design of alarm circuits.	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the concepts of automatic Control with examples.**		
M1.01	Define the concepts of process control.	3	Understanding
M1.02	Explain automatic control	3	Understanding
M1.03	Explain the process and controller characteristics	4	Understanding
M1.04	Describe the different process control loops	5	Understanding
Contents: Process- Process Control and Process Plant -Human aided control and Automatic control- Block diagram of process control - Error, set point, controlled variable, manipulated variable and measured variable -Elements of process control loop - Process characteristics - Process equation, Process Load, Process Lag and Selfregulation, Control System Parameters - Error, Variable Range, Control Parameter Range, Control Lag - Dead Time, Cycling - Process Parameters - Temperature process control , Pressure Process control system - Flow process control system , Level process control system.			
CO2	**Illustrate controller tuning methods.**		
M2.01	Describe different types of process	4	Understanding
M2.02	Explain different discontinuous controller Modes	3	Understanding
M2.03	Explain different continuous controller Modes	4	Understanding
M2.04	Illustrate tuning of controller	3	Applying
	Series Test - I	1	

Contents:

Batch and Continuous process - Discontinuous control Modes - two position - neutral zone -multi position control Mode- Continuous Control Modes - proportional control Mode -proportional band and offset error - integral control Mode – reset rate -derivative control Mode - derivative time- composite control Modes- PI, PD and PID control Modes –Electronic Controllers-Electronic Error detector-Electronic P,PI,PD and PID controller using Op -amp- Tuning methods – open loop method - Ziegler Nichols method.

CO3	**Explain the concepts of final control operation.**		
M3.01	Describe the final control operation	3	Understanding
M3.02	Explain the working of different actuation methods	3	Understanding
M3.03	Explain the different control valve types according to actuation and port design.	4	Understanding
M3.04	Describe the control valve accessories	4	Understanding

Contents:

Final control element -block diagram of final control operation - working of pneumatic, electric and hydraulic actuators - Control Valves –Important parts of control valve with schematic diagram-air to open and air to close control valves - Different valve plugs – Single seated and Double seated valve- Butterfly valves, Ball valve, Globe valve – Electrically actuated valves - Motor operated valve and Solenoid valve -Inherent flow characteristics- Control Valve coefficient Cv, Rangeability-Control Valve Sizing-Cavitation and flashing-Auxiliary units for control valve – valve positioner, position transmitter, limit switch, booster relay, air pressure regulator - I/P converter.

CO4	**Design of alarm circuits.**		
M4.01	Design alarm and annunciator circuits using basic digital gates and comparators.	3	Applying
M4.02	Describe the different telemetry systems	4	Understanding
M4.03	Explain the concept of Foundation fieldbus andProfibus DP	4	Understanding
M4.04	Explain the concept of HART communication protocol	4	Understanding
	Series Test - II	1	

Contents:

Alarms and Annunciators - Role and functions of Alarms – Implementation of Alarms using logic gates and comparators-Telemetry system- General Telemetry system block Diagram- Voltage Telemetry System- Current Telemetry System - Motion balance and Force balance telemetry systems - Position Telemetry System- Digital communication channels - Field bus- advantages of field bus- Profibus DP -Functional elements in Foundation Field bus. Highway –Addressable Remote Transducer-Benefits of HART field communication protocol. Working of HART - Block diagram of HART digital communication System - HART Specification (Layers of HART).

Text / Reference:

T/R	Book Title/Author
R1	Curtis. D. Johnson, Process control instrumentation technology, PHI.
R2	Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company.
R3	Krishnakanth, Computer based Industrial Control, PHI.
R4	Donald .P. Eckman, Automatic process control, WILLEY India.
R5	Ernest O Doebelin, Measurement Systems application and Design 5th Edition TMH.

Online Resources:

Sl.No	Website Link
1	http://nptel.ac.in/courses/103103037/
2	https://www.youtube.com/watch?v=U-uUYCFDTrc
3	https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf
4	https://processcontrol.tv/tutorials
5	https://www.pidlab.com/en/virtual-labs